Literature Review: Deep Learning for Seizure Detection and Prediction from Non-EEG Wearable Signals (2015–2025)(Exposé)

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Contents

1 Introduction

1.1 Problem Statement and Motivation

Epilepsy, a chronic neurological disorder affecting 60 million people wordlwide, with approximately one-third of patients remaining drug-resistant (Hussein et al., 2018). Sveinsson et al., 2020 found that 69% of Deaths due to a Generalized tonic-clonic seizure could have been prevented if not left unattended, highlighting the demand for reliable and timely prediction.

While reliable prediction is already possible through electroencephalography (EEG), its operational complexity render it unsuitable for ambulatory use. To address the need for ambulatory detection many are experimenting with utilizing autonomic and motion signals to detect and predict seizures.

However, translating and analysing these often noisy and complex time-series physiological signals has pushed traditional machine learning (ML) approaches to their limits. These approaches require manual engineering such as cleaning and extracting of the data, often failing to capture the intricate and non-linear complex temporal dynamic of these signals. It is this limitation that deep learning (DL) models are designed to overcome, with their ability of hierarchical and learned feature extraction uniquely addressing these limitations.

This review therefore synthesizes and critically evaluates recent advances in time-series deep learning—particularly convolutional neural networks (CNNs) and long short term memory (LSTMs)— applied to seizure detection and prediction using non-EEG wearable biosignals. Central to this analysis is the question of how architectural choices, training paradigms (e.g., transfer learning, personalization), and rigorous, prospective evaluation can converge to meet the stringent clinical requirements of high sensitivity and minimal false alarm rates (FAR) in real-world deployment.

2 Theoretical Foundations for Ambulatory Seizure Monitoring

2.1 The Ambulatory Monitoring Paradigm: Requirements and Challenges

Beniczky et al., 2021; Chen et al., 2022 found that translation of seizure detection from the controlled clinical environment to an ambulatory setting redifines approaches and performance requirements. In clinical environments maximum seinsitivity is prioritized over FAR, when engineering for ambulatory use a balance of both metrics is needed. Failing to detect a seizure (low sensitivity) poses a safety risk, While excessive false alarms (high FAR) leads to desensitizing and "alarm fatigue" and the device being ignored or not used at all. Users reported that devices should achieve high sensitivity ($\geq$90%) with low false alarm rates — acceptable FARs were 1-2/week for patients with $\geq$1 seizure/week and 1-2/month for patients with <1 seizure/week (Sivathamboo et al., 2022). While more complex models can find more intricate patterns, they also introduce the "black-box" problem wherein the model lacks transparency AbuAlrob et al., 2025; Ghaderi, 2025. Therefore, the design should incorporate explanations for decisions to ensure user trust and regulatory compliance AbuAlrob et al., 2025; Miron et al., 2025.

2.2 Physiological Biosignals for Wearable Sensing

Wearable devices circumvent the impracticality of EEG by capturing peripheral biosignals that serve as proxies for central nervous system activity during seizures. These proxies will inherently relay incomplete and noisy signals so a combination of multiple sensors is required. The acceleration/attitude sensor excels at detecting muscle convulsions but is noisy in daily use, as it captures all movement. The electrodermal activity (EDA) sensor detects autonomic arousal by measuring the skin conductance associated with sweat gland activity.

2.3 Seizure Manifestations and Corresponding Biosignals

Wearable biosignals act as peripheral proxies for cerebral activity during seizures. Different seizure types produce distinct physiological signatures across these signals, informing detection approaches. This relationship centers on two primary manifestations: motor activity and autonomic activation.

Convulsive seizures, such as generalized tonic-clonic events, produce clear motor signatures captured by accelerometry (ACC), which measures movement and posture changes characteristic of ictal events (Wang et al., 2025; Wu et al., 2024).

Simultaneously, seizures drive autonomic responses recorded through complementary biosignals. Electrodermal activity (EDA) measures sympathetic nervous system arousal via skin conductance changes (Jain et al., 2017; Miron et al., 2025). Cardiac activity provides particularly strong autonomic proxies: electrocardiography (ECG) captures heart rate and rhythm alterations (Ghaderi, 2025; Leal et al., 2017), while heart rate variability (HRV) quantifies autonomic balance dynamics (Mason et al., 2024; Pavei et al., 2017).

In wearable applications, photoplethysmography (PPG) serves as a practical alternative for cardiac monitoring, deriving pulse rate and variability though with greater motion sensitivity than ECG (Chen et al., 2022; Seth et al., 2023).

Consequently, detection strategies combine these biosignals according to target seizure types, with multimodal fusion providing the most robust approach for ambulatory monitoring.

2.4 Evaluation Frameworks: From Retrospective Analysis to Prospective Validity

The clinical validity of a seizure detection algorithm is determined by the rigor of its evaluation, formalized in the ILAE's phased framework (Phases 0–4) (Beniczky et al., 2021). Retrospective studies (Phases 0-2), while foundational, risk significant data leakage and optimistic bias if they use non-chronological data splits, allowing future information to inform training (Kalousios et al., 2024; Wong et al., 2023). Pseudo-prospective validation addresses this by enforcing chronological order, training only on past seizures to test on subsequent ones, providing a more realistic performance estimate (Kalousios et al., 2024; Lu et al., 2025). The clinical gold standard is a prospective trial (Phase 3), which requires a locked algorithm, a video-EEG reference standard, and testing on new, unseen data from multiple centers. The final step is in-field evaluation (Phase 4) in patients' homes, where metrics like the false alarm rate per hour (FAR/h) become decisive for practical usability, as demonstrated by long-term studies like Nightwatch (Miron et al., 2025; Shum & Friedman, 2021).

3 Methodological Approach and Structure

The review follows structured IS/medical literature guidance (concept matrix inspired by (Webster & Watson, 2002) and PRISMA principles). The methodological approach uses a multi-stage search and transparent extraction/synthesis procedures as outlined below.

1. **Search strategy (Multi-stage, 2015–2025):**
    - *Scoping:* Google Scholar for broad coverage and identifying candidate papers.

- *Formal queries:* IEEE Xplore, PubMed, and Scopus using controlled keyword strings and deduplication.
- *Citation chasing:* Backward and forward citation tracking on key papers to ensure completeness.

2. **Inclusion / Exclusion criteria:**

   - *Inclusion:* Studies using wearable/non-EEG autonomic signals for seizure detection or preictal prediction.
   - *Exclusion:* EEG-only studies, invasive recordings, or papers limited to purely algorithmic simulations without human data.

Extracted dimensions populate a concept matrix with the following fields:

1. Setting: clinical vs ambulatory.
2. Modality: ECG/HRV, PPG, EDA, ACC.
3. Task: detection vs prediction.
4. Model class and fusion strategy.
5. Study phase: retrospective, pseudo-prospective, prospective.
6. Evaluation granularity: sample- vs alarm-based.
7. Windowing specifics (window length, stride, overlap).
8. Metrics: sensitivity, FAR/h, AUC, f1-score, latency.
9. Personalization.
10. Interpretability and safety considerations.
11. Data characteristics and dataset identity.
12. Missing data handling (imputation, exclusion) and synthetic data use.
13. Inference requirements (latency, computational load, hardware).

Prioritization Criteria: Studies including splits preserving patient-specific data, FAR/h reporting and external or cross validation will be prioritized in the synthesis. A PRISMA-style flow diagram will summarize and visualize screening and inclusion.

4 Modeling Landscape and Design Dimensions

Detection and preictal prediction from ECG/HRV, PPG, EDA, and ACC are framed by windowing, labeling, and alarm logic, with alarm-based sensitivity, FAR/h, and latency as the outcomes used in the research questions.

4.1 Architectural Evolution: Feature Engineering vs. Deep Learning

Classical pipelines utilizing HRV, EDA, and ACC features combined with SVM, RF, or kNN remain competitive in retrospective settings (Mason et al., 2024; Seth et al., 2023). However, deep temporal models such as 1D-CNN, LSTM, and TCN often improve phase discrimination. Recently, attention variants and Transformers are being explored on these signals (Abtahi et al., 2025; Kalousios et al., 2024; Pordoy et al., 2025; Yang et al., 2022). A key inquiry of this review is determining how feature-based models compare to deep models and whether attention mechanisms add distinct value under ambulatory, alarm-based evaluation.

4.2 Sensor Modalities and Fusion Strategies

Fusion across ECG, EDA, ACC, and PPG at the feature, decision, or representation level seeks complementary cues and is relevant to maintaining performance under noise and non-stationarity (Pordoy et al., 2025; Yang et al., 2022).

4.3 Personalization and Domain Shift

Patient-specific fine-tuning and post-hoc calibration shift the operating point toward lower FAR/h and higher sensitivity in ambulatory use (Andrade et al., 2024). Thresholding and periodic recalibration are commonly used to handle drift.

4.4 Robustness and Generalization

Robustness requires explicit checks of domain shift across patient, device, site, and activity. Cross-dataset evaluation (e.g., training on TUH and testing on RPAH or EPILEPSIAE) exposes this shift and favors alarm-based over sample-based metrics (Andrade et al., 2024; Yang et al., 2022). Artifact handling, augmentation through time-warping, jitter, or scaling, and Out-of-Distribution (OOD) detection support generalization (Kalousios et al., 2024; Seth et al., 2023).

4.5 Deployment Constraints: Latency, Energy, and Interpretability

Wearable seizure detection imposes strict deployment constraints. Detection latency must remain below clinically relevant thresholds (typically seconds to allow caregiver response or patient warning), which limits model complexity and inference time. Energy consumption determines battery life; continuous inference on resource-constrained MCUs or mobile processors requires lightweight architectures such as quantized CNNs or efficient RNNs. Compact sequence models (e.g., depthwise-separable 1D-CNNs, temporal convolutional networks) and event-driven or on-demand processing reduce computational burden.

Interpretability remains critical for clinical adoption. Regulators and clinicians require transparency in alarm generation to trust automated decisions affecting patient safety. Post-hoc explanation methods (e.g., attention weights, Grad-CAM on temporal segments, SHAP values for HRV features) can highlight which signal segments or features triggered an alarm. Hybrid pipelines that combine interpretable HRV/EDA features with deep representations offer a middle ground between performance and explainability.

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